A Hybrid Bi-LSTM and Ensemble Framework for Enhanced Hate Speech Detection

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Course: CSE424

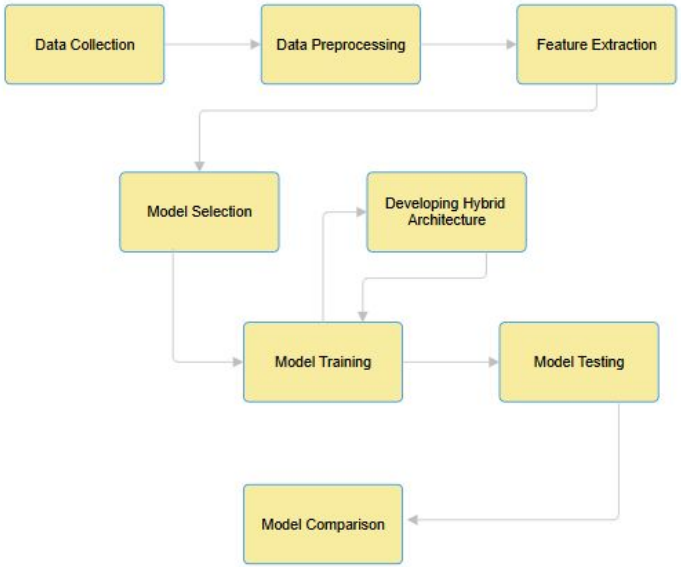
AIM OF THIS STUDY

The rapid Growth of Social media necessitates for automated detection of Hate-speech targeted towards people and groups. The Primary aim of this research is to explore various machine learning algorithms and develop a **hybrid ensemble architecture** using **deep learning models** and **ensemble learning models**.

Research Questions

- How was the baseline performances for machine learning models for this task?
- How did deep learning models perform in comparison to the baseline models?
- Which models performed the best for this task?
- Can Hybrid Architectures be built for the task while combining the best performing models and deep learning models using ensemble techniques?

Methodology



The research pipeline followed was Data Collection (Crowdflower Dataset [1]), Exploratory Data Analysis, Data Preprocessing, Feature extraction, Data Splitting (80% Training, 20% Test data), Model Selection & model training.

Based on the results of the Models on the training set, models were chosen for the Hybrid architecture, then the hybrid architectures were trained and finally all the models were tested and compared.

Data Preprocessing

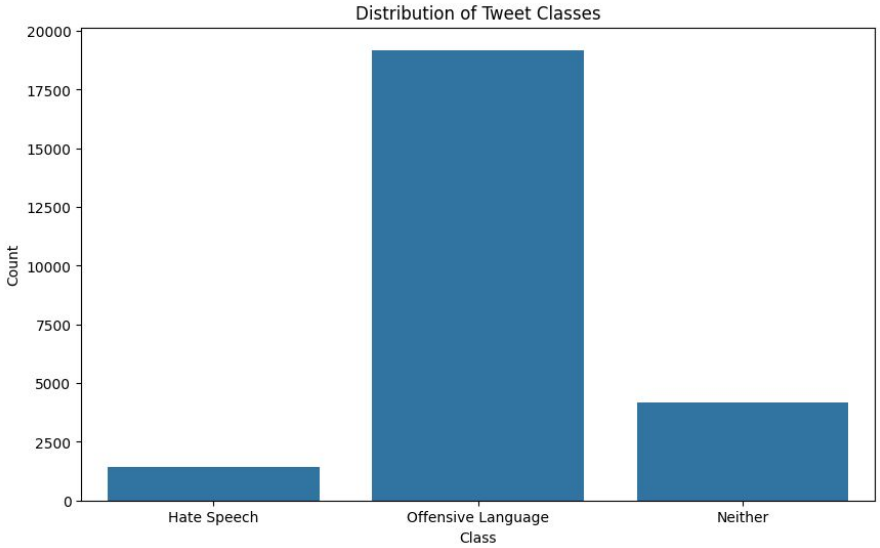
The Crowdflower (Davidson et al., 2017) dataset has been used which has 24,783 Rows, 7 columns, 0 null and 0 duplicate values

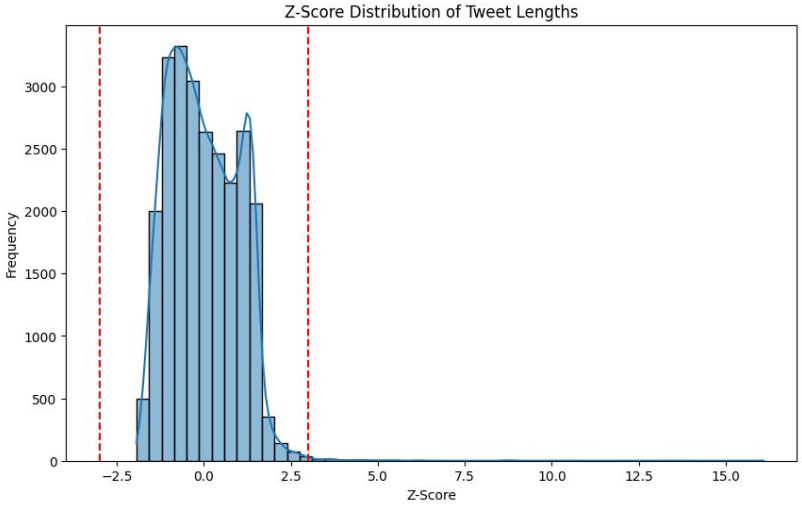
The Preprocessing steps include:

- Removing punctuation
- Outlier Removal
- Resampling
- Removing stopwords
- Tokenization
- Lemmatization.

The preprocessing steps were crucial for extracting deeper structural features like sentiment shifts and slang markers.

Issues in EDA: Class imbalance and Text length outliers





class		
Offensive Language	19190	77.432111
Neither	4163	16.797805
Hate Speech	1430	5.770084

The tweet length outliers were dealt with using Z-score |<3.5|; and then the issues faced in class imbalance was mitigated by resampling the offensive_language class, which had 19190 tweets, to 9,000 tweets and upsampled the hate_speech class from 1430 to 2858.

Feature Extraction

1. Bert Embeddings
2. TF-IDF Matrix:
3. N-gram Matrix:
4. Numeric Features:
   a. Sentiment Score
   b. Readability Score
   c. Slang marker
   d. NER
   e. Mention (@) + aggressive verb count

   .

   .

A total of 17 of these numeric features were extracted from text, then scaled and grouped together.

Then SMOTE was applied to deal with the class imbalance.

For training the models, dataset was split in 80-20 for training and testing.

Traditional ML Models used

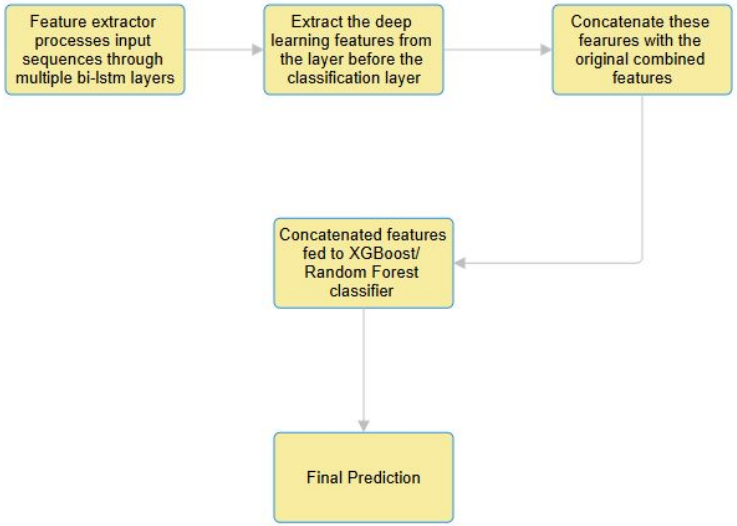
- Naive Bayes: to establish a performing baseline
- Logistic Regression: to see how linear decision boundary classifier works for the dataset
- Random Forest: to establish the performance baseline for ensemble classifiers
- XGBoost: further performance analysis on another ensemble classifier

Deep Learning models used

- Bi-LSTM: to set a performance baseline for deep learning methods.

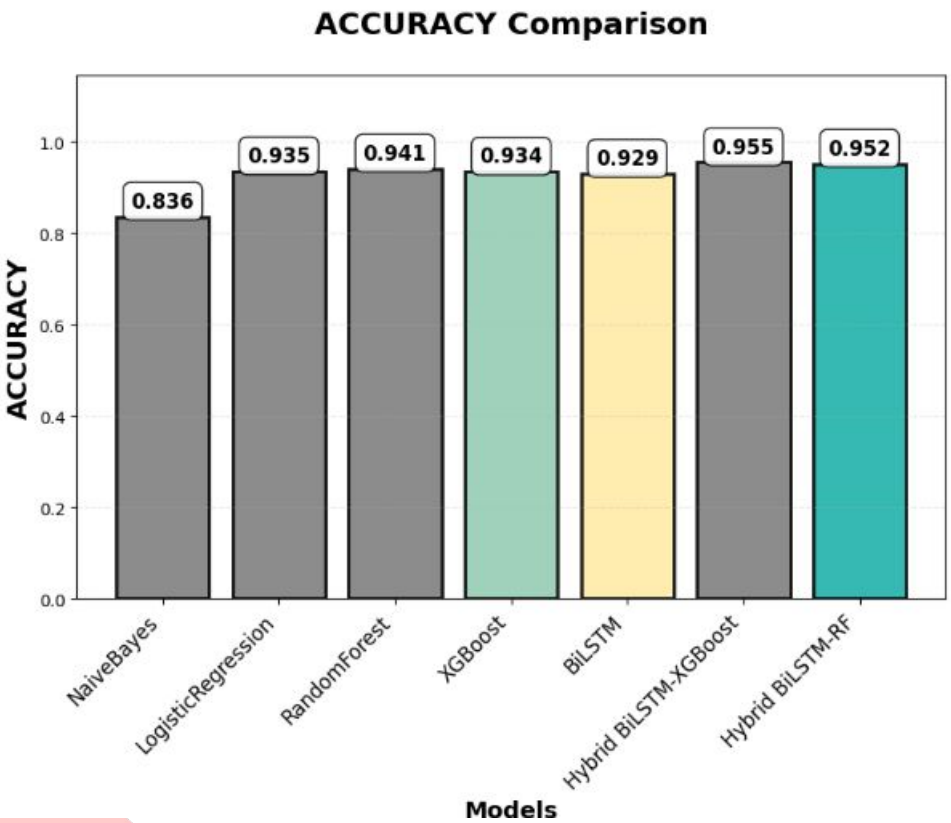
Hybrid Models Proposed

- Bi-LSTM + XGBoost
- Bi-LSTM + Random Forest



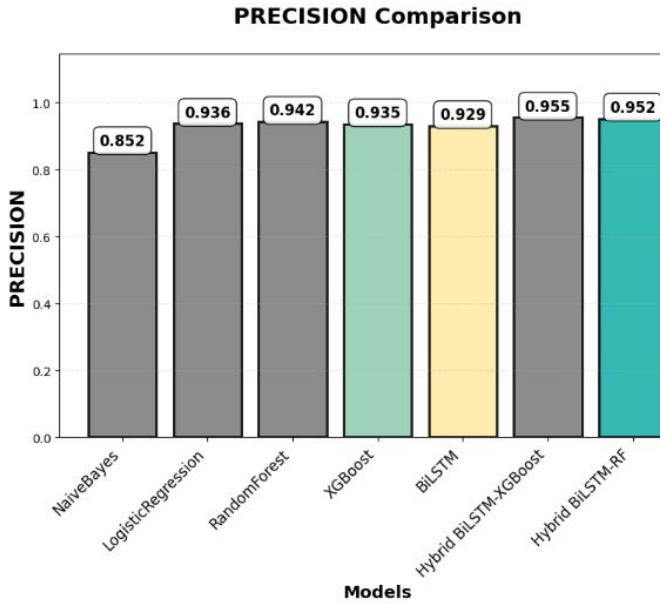
The pipeline was followed for both the hybrid architectures, the learning from the 4 layers of B-LSTM is combined with the ensemble classifier for superior performance by learning from scaled 7851 dimensional features and better interpretability from ensemble techniques.

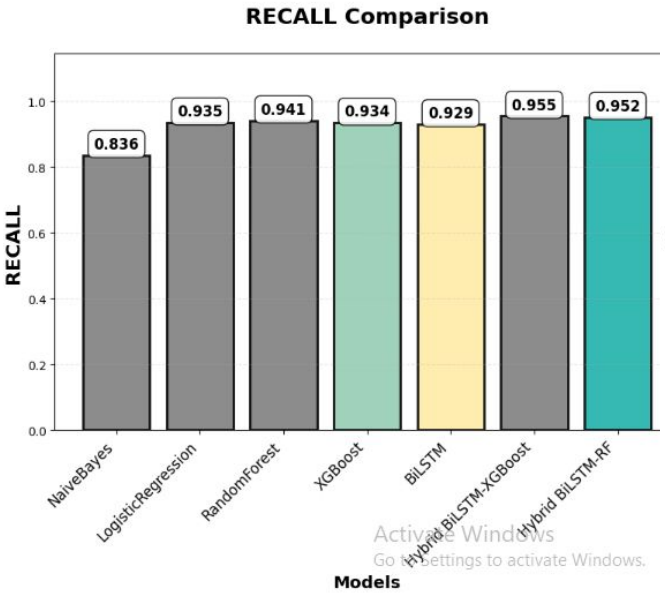
Evaluation Across All Models

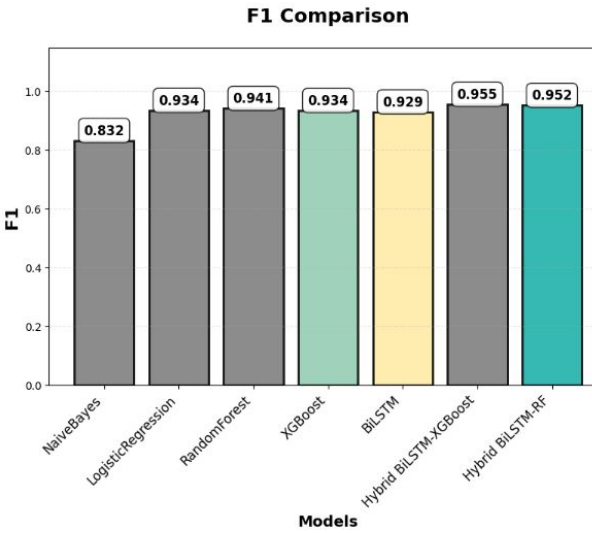


Model	Accuracy	Precision	Recall	F1-Score	AUC
Naïve Bayes	0.8359	0.8520	0.8359	0.8322	0.8783
Logistic Regression	0.9352	0.9364	0.9352	0.9344	0.9764
Random Forest	0.9411	0.9423	0.9411	0.9413	0.9904
XGBoost	0.9344	0.9354	0.9344	0.9340	0.9878
Bi-LSTM	0.9294	0.9293	0.9294	0.9293	0.9865
Hybrid BiLSTM-XGBoost	0.9550	0.9549	0.9550	0.9550	0.9915
Hybrid BiLSTM-Random Forest	0.9517	0.9516	0.9517	0.9516	0.9922

Evaluation Across All Models







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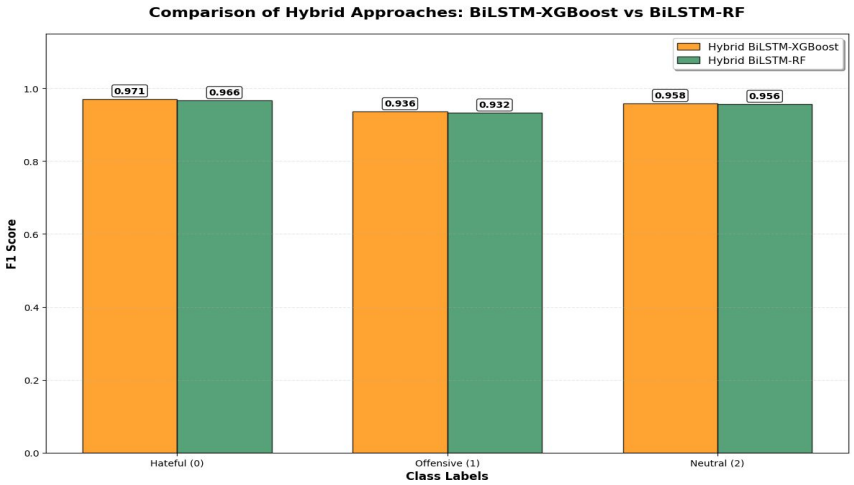
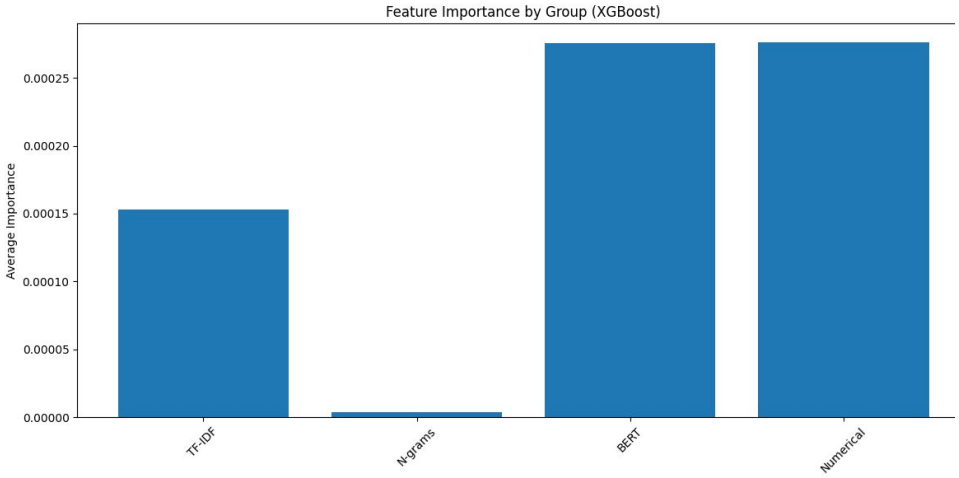
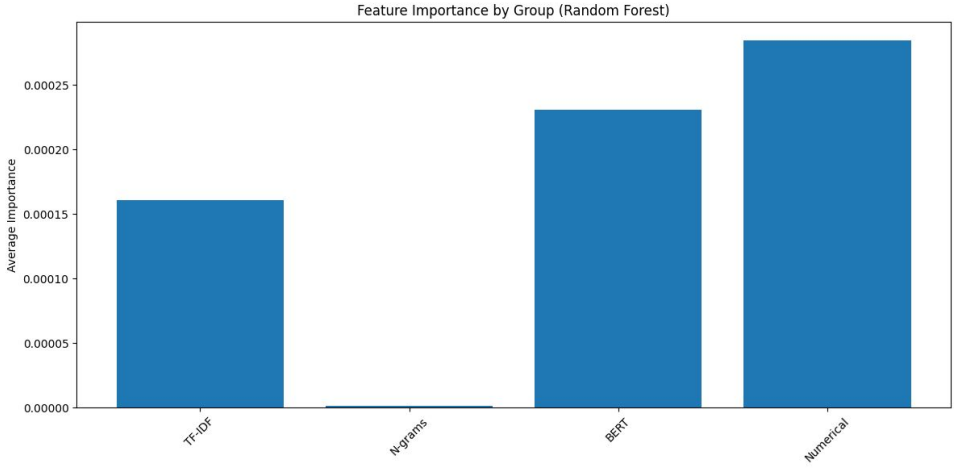
F1-Scores by Class Comparison: Hybrid Models

Bi-LSTM + XGBoost outperforms the other models with an accuracy score of **95.50%,** and has a slightly better advantage over Bi-LSTM + Random Forest of about **0.34% in F1 Scores.** Overall, both the hybrid models perform exceptionally well.



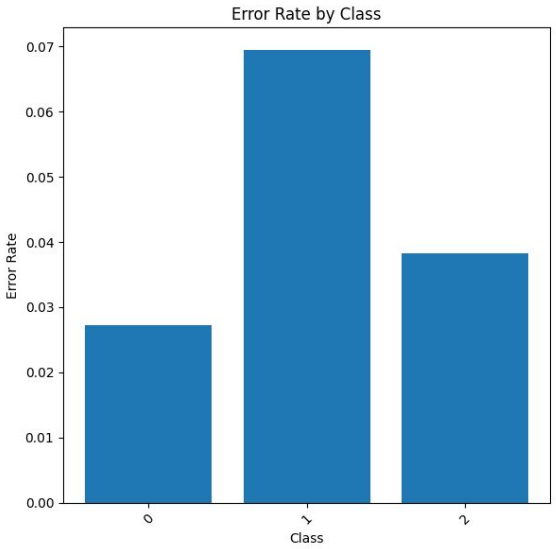
Feature importance: Hybrid Models

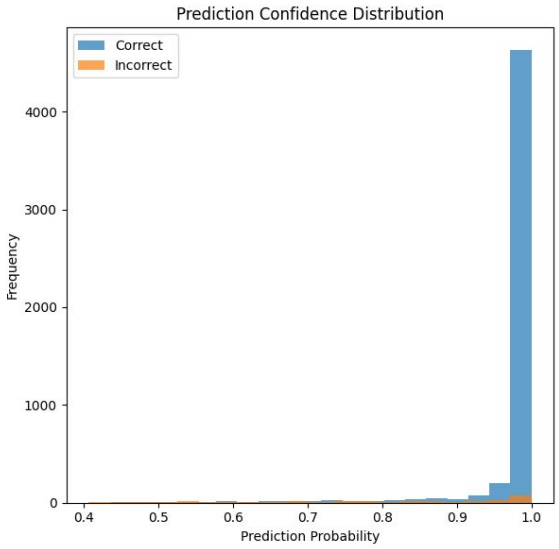


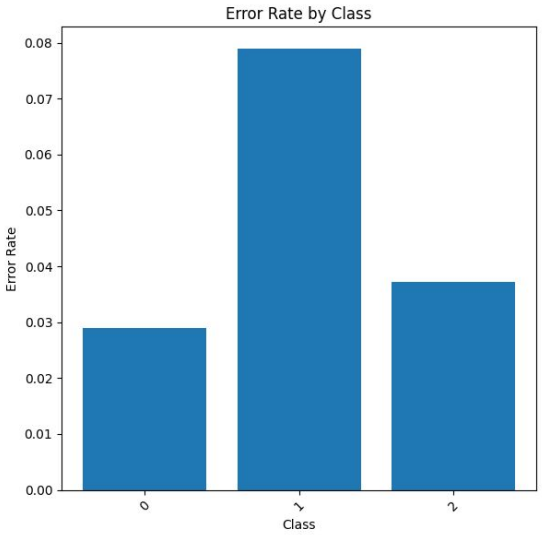


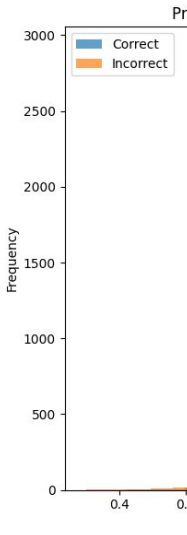
Feature importances of the Hybrid Models show how well the extracted Numeric features contributed to the classification; alongside the contextual embeddings from deep learning.

Error Rates: Hybrid Models



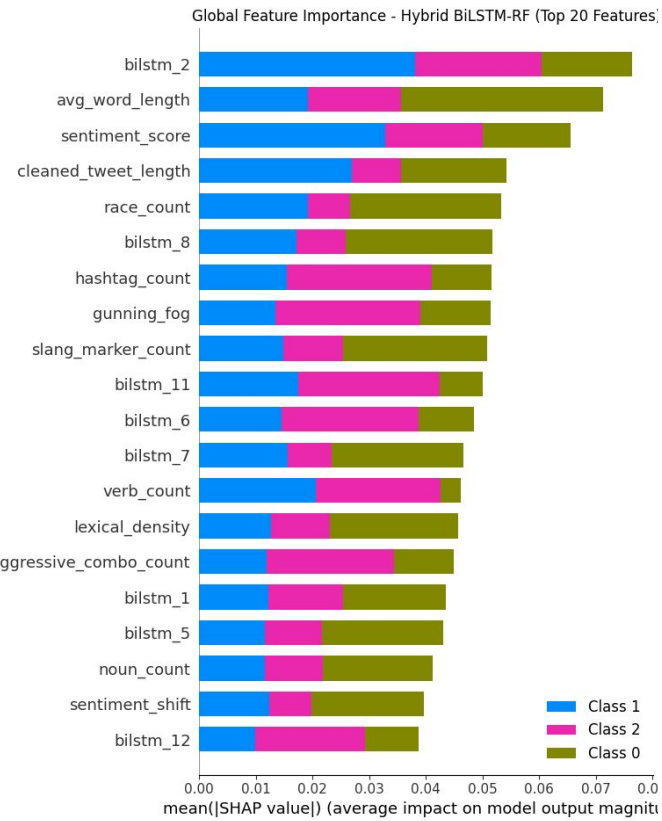


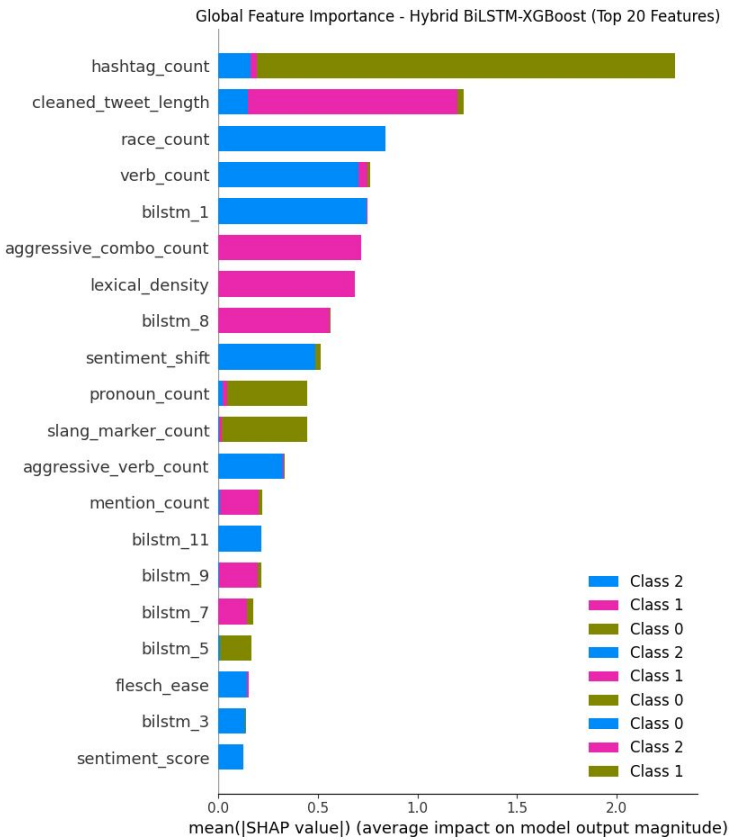




Error Rates indicate the models still have issues regarding class 1 (‘offensive_language’), even though the error rate for class 0 (‘hate-speech’) is the lowest for both, indicating a success in one of the aims of the research.

Explainable AI: SHAP





The SHAP analysis provides the necessary interpretability for the hybrid models; where the Bi-LSTM - RF model learned the most from **bi-lstm embeddings** while still having a healthy balance between lexical features. Contrarily, Bi-LSTM - XGB learned the most from **metadata features** as well as having a balanced feature impact from the extracted features as well.

Future Work

- · Integration with datasets from other platforms (such as chat-logs, comment sections from various public servers from Facebook, Instagram, Discord, Reddit, Quora etc.), larger as well as multimodal datasets combining image and sound data, as social media platforms nowadays are endlessly expandable with audio-visual data dominating the scenario. Furthermore, including different forms of hate speech such as 'Harassment', 'Bullying', 'Racism', 'Sexism', 'Xenophobia', 'Bigotry', 'Fascism' etc. as well as being able to classify non-consensual advances such as different threats of 'Predatory', 'Pedophilic', 'Sexual Assault' etc. for better identification of the type of offense that is being threatened in the speech.
- · Exploration of other deep learning models like GRU, RNN, CNN and transformers like BERT and different versions of it like RoBERT, DistilBIRT and seeing how they fair if combined with similar ensemble techniques.
- · Improved feature extraction for better distinguishability between 'Hateful' and 'Offensive', such as implementing LDA, other contextual embeddings (Word2Vec, GloVe etc.), other metadata features like tweet timing, reply velocity etc. as well as additional hyperparameter tuning to make sure the model gets the most from deep learning.

Thank You!